

Plant classification

THE PLANT KINGDOM is divided into different groups and contains about 400,000 separate species that we know about. The majority of plant organisms belong to the flowering plant family, or angiosperms. Plants that have seeds but no flowers are called gymnosperms. The groups shown here cover all of the main plant classifications.



Ginkgo

GINGKO
The ginkgo, or maidenhair, tree is native to China. A gymnosperm, it is related to conifers, but it has many unusual features, which is why it is classified in a group of its own. Unlike most conifers, ginkgos are deciduous and have fan-shaped leaves.

Most liverworts have leaves, but some are flat and leafless



Liverwort

MOSSES AND LIVERWORTS

Mosses and liverworts belong to a group called the bryophytes, which number 14,000 species. These small plants usually grow in shaded, damp places. They first appeared around 425 million years ago, and they contributed to the formation of coal and peat.

FLOWERING PLANTS

Flowering plants, or angiosperms, constitute about 250,000 species—the vast majority of plants. A flower is a specialized part of the plant that develops into a fruit, which contains one or more seeds housed in ovaries. These seeds must be pollinated and dispersed to ensure the continuation of each species. There are two types of flowering plant: monocots and dicots.

Monocots have a single cotyledon, or seed-leaf. Their adult leaves are often long and narrow, with parallel veins. Monocots include cereals, such as wheat, some vegetables, such as leeks, some fruits, such as pineapples, and also orchids and lilies. They number about 55,000 species.

Dicots are plants whose seeds have two cotyledons, or seed-leaves. Their adult leaves usually have a network of veins around a central midrib. Dicots total at least 200,000 species, and they include most shrubs, and all the world's broadleaved trees.

Wheat



Pineapple



Leek

Rose



Dicot plants often have woody stems



Cabbage



Cactus